```
'Ke'Shawn Stafford
'Project 11
'05/01/2025

' {$STAMP BS2}
' {$PBASIC 2.5}

x         VAR Word
x1        VAR Word
y         VAR Word
y1        VAR Word
ang_x     VAR Word
ang_y     VAR Word
counter   VAR Word
counter1  VAR Word
pos       VAR Word
ball      VAR Word
LR        VAR Word
UD        VAR Word

SEROUT 14, 84, [22,12,17]
GOSUB map
DO
pos = 157
ball = 7
PULSIN 10, 1, x
PULSIN 11, 1, y
x = x / 15
y = y / 15
x1 = x
y1 = y
PAUSE 50
GOSUB map
GOSUB servo
GOSUB x_pos
GOSUB y_pos
SEROUT 14, 84, [pos, ball]
LOOP

servo:
      y1 = y1 - 126
      y1 = y1 * 12
      y1 = y1 + 250
      PULSOUT 12, y1
      PAUSE 20
      x1 = x1 - 130
      x1 = x1 * 12
      x1 = x1 + 250
      PULSOUT 15, x1
      PAUSE 20
      RETURN

angle:
RETURN

x_pos:
      IF (x = 169 OR x = 168) THEN
      LR = 0
      counter = 1
      ELSEIF (x = 167 OR x = 166) THEN
      LR = -1
      counter = 2
      ELSEIF (x < 166 AND x > 161) THEN
      LR = -1
      counter = 1
      ELSEIF (x < 161 AND x > 153) THEN
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LR = -2
IF (x > 157) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x < 154 AND x > 146) THEN
LR = -3
IF (x > 149) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x < 147 AND x > 138) THEN
LR = -4
IF (x > 141) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x < 139 AND x > 129) THEN
LR = -5
IF (x > 133) THEN
counter = 2
ELSE
counter = 1
ENDIF

ELSEIF (x = 170) THEN
LR = 0
counter = 3

ELSEIF (x = 171 OR x = 172) THEN
LR = 0
counter = 2
ELSEIF (x = 173 OR x = 174) THEN
LR = 1
counter = 1
ELSEIF (x > 174 AND x < 183) THEN
LR = 1
IF (x > 177) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x > 182 AND x < 191) THEN

LR = 2
IF (x > 185) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x > 190 AND x < 199) THEN
LR = 3
IF (x > 194) THEN
counter = 2
ELSE
counter = 1
ENDIF
ELSEIF (x > 198 AND x < 207) THEN
LR = 4
IF (x > 203) THEN
counter = 2
ELSE
counter = 1
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      ENDIF
      ELSEIF (x = 207 OR x = 208) THEN
      LR = 5
      counter = 1
      ELSEIF (x = 209 OR x = 210) THEN
      LR = 5
      counter = 2
      ENDIF
      pos = pos + LR
      RETURN
y_pos:
      IF (y = 166) THEN
      UD = 0
      IF counter = 2 THEN
      ball = 4
      ELSEIF counter = 1 THEN
      ball = 3
      ELSEIF counter = 3 THEN
      ball = 7
      ENDIF
      ELSEIF (y = 167) THEN
      IF counter = 1 THEN
      ball = 5
      ELSE
      ball = 6
      ENDIF
      ELSEIF (y < 165 AND y > 145) THEN
      IF counter = 1 THEN
      ball = 1
      ELSE
      ball = 2
      ENDIF
      ELSEIF (y > 167 AND y < 182) THEN
      UD = 20
      IF (y > 167 AND y < 173) THEN
      IF counter = 1 THEN
      ball = 1
      ELSE
      ball = 2
      ENDIF
      ELSEIF (y > 172 AND y < 177) THEN
      IF counter = 1 THEN
      ball = 3
      ELSE

      ball = 4
      ENDIF
      ELSEIF (y > 176 AND y < 182) THEN
      IF counter = 1 THEN
      ball = 5
      ELSE
      ball = 6
      ENDIF
      ENDIF
      ELSEIF (y > 181 AND y < 195) THEN
      UD = 40
      IF (y > 181 AND y < 188) THEN
      IF counter = 1 THEN
      ball = 1
      ELSE
      ball = 2
      ENDIF
      ELSEIF (y > 187 AND y < 194) THEN
      IF counter = 1 THEN
      ball = 3
      ELSE
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      ball = 4
      ENDIF
      ELSEIF (y > 193 AND y < 200) THEN
      IF counter = 1 THEN
      ball = 5
      ELSE
      ball = 6
      ENDIF
      ENDIF

      ELSEIF (y < 145) THEN
      UD = -20
      IF (y > 138 AND y < 145) THEN
      IF counter = 1 THEN
      ball = 5
      ELSE
      ball = 6
      ENDIF
      ELSEIF (y > 132 AND y < 139) THEN
      IF counter = 1 THEN
      ball = 3
      ELSE
      ball = 4
      ENDIF
      ELSEIF (y < 132) THEN
      IF counter = 1 THEN
      ball = 1
      ELSE
      ball = 2
      ENDIF
      ENDIF
      ENDIF
      pos = pos + UD
      RETURN

map:
      SEROUT 14, 84, [248,
      %00100,
      %00100,
      %00100,
      %00100,
      %00100,
      %00100,
      %11111,
      %00000]
      SEROUT 14, 84, [249,
      %00000,
      %00000,
      %01000,
      %00000,
      %00000,
      %00000,
      %00000,
      %00000]
      SEROUT 14, 84, [250,
      %00000,
      %00000,
      %00010,
      %00000,
      %00000,
      %00000,
      %00000,
      %00000]
      SEROUT 14, 84, [251,
      %00000,
      %00000,
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%00000,
%00000,
%01000,
%00000,
%00000,
%00000]
SEROUT 14, 84, [252,
%00000,
%00000,
%00000,
%00000,
%00010,
%00000,
%00000,
%00000]
SEROUT 14, 84, [253,
%00000,
%00000,
%00000,
%00000,
%00000,
%00000,
%01000,
%00000]
SEROUT 14, 84, [254,
%00000,
%00000,
%00000,
%00000,
%00000,
%00000,
%00010,
%00000]
SEROUT 14, 84, [255,
%00000,
%00000,
%00000,
%00000,
%00100,
%00000,
%00000,
%00000]
SEROUT 14, 84, [152, "_", "_", "_", "_", "_", 0, "_", "_", "_", "_", "_", "_"]
SEROUT 14, 84, [132, " ", " ", " ", " ", " ", " ", "|", " ", " ", " ", " ", " ", " "]
SEROUT 14, 84, [172, " ", " ", " ", " ", " ", " ", "|", " ", " ", " ", " ", " ", " "]
SEROUT 14, 84, [192, " ", " ", " ", " ", " ", " ", "|", " ", " ", " ", " ", " ", " "]
RETURN
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